

Course: ENSF 694

Final Project

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1: Introduction

2: Design decisions

Design Decisions: For each required feature, describe:

- The data structure or algorithm you chose to implement it.
- Why this choice is appropriate for the problem (access patterns, insertion frequency, ordering requirements, etc.).
- At least one alternative you considered and why you rejected it.

2.1: Campus Map and Shortest Path Navigation

2.2: Route History and Navigation Undo

2.3: Room and Event Booking System

2.4: Priority-Based Service Queue

2.5: Fast Building and Resource Lookup

2.6: Incoming Request Processing

2.7 (Bonus) Balanced Event Index

3: Complexity Analysis

Complexity Analysis: For each major operation in your system, provide:

- Time complexity (best, average, and worst case where applicable).
- Space complexity.
- A brief justification for each bound.

4: Challenges and Lessons Learned

5: Contributions

Eric	Dylan	Zohara

6: Demos

3.3 Demo Scenarios and Screenshots

Your report must include a dedicated section with screenshots of your running application, organized by scenario. Include at minimum the following use cases:

- Shortest path query: Show at least two route queries with different source/destination pairs, displaying the full path and total cost.
- Undo navigation: Show a sequence of route queries followed by one or more undo operations, demonstrating correct state restoration.
- Booking range query: Show a time-range query returning all bookings within a specified window.

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- Priority queue demo: Show requests being inserted at different priority levels, then dequeued in correct priority order.
 - Fast lookup demo: Show insert and lookup operations on the building/resource store, including a lookup for a key that does not exist.
 - Request pipeline demo: Show at least 10 requests being enqueued and dequeued in arrival order.

Each screenshot must be accompanied by a one-to-two sentence caption explaining what is being demonstrated and which feature/component is in use.

6.1: Shortest Path Query

6.2: Undo Navigation

6.3: Booking Range Query

6.4: Priority Queue Demo

6.5: Fast Lookup Demo

6.6: Request Pipeline Demo